

Yang Yan

Ph.D. Candidate, Zhejiang University / Westlake University

AI for Structural Biology | Scientific Foundation Models | Computer Vision | Representation Learning

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Research Profile

AI-for-science Ph.D. candidate developing foundation models, representation learning methods, and scalable ML systems for structural biology, where cryo-electron microscopy (cryo-EM) provides noisy 2D images used to reconstruct 3D molecular structures. First author of a *Nature Methods* study introducing **Cryo-IEF**, a self-supervised foundation model pretrained on **65.3M cryo-EM particle images** used for 3D molecular structure reconstruction and transferred to classification, clustering, quality ranking, and automated reconstruction. My work connects large-scale visual pretraining, scientific data infrastructure, and domain-grounded automation to help turn complex molecular imaging data into reliable structural models with less manual expert intervention.

Education

2023–Present	Ph.D. Candidate in Computer Science Joint Ph.D. program supervised by Prof. Fajie Yuan and Prof. Huaizong Shen. Research: scientific foundation models, large-scale representation learning, automated reconstruction systems, and AI for structural biology.	Zhejiang University / Westlake University
2020–2023	M.S. in Translational Medicine (Engineering) National Institute of Diagnostics and Vaccine Development in Infectious Diseases, supervised by Prof. Ningshao Xia. Research: machine learning for medical image analysis and cryo-EM image representation.	Xiamen University
2016–2020	B.S. in Electrical Information Engineering Research included machine-learning applications in localization and signal analysis.	Northeastern University (Qinhuangdao)

Selected Publications

A comprehensive foundation model for cryo-EM image processing 2026

Yang Yan, Shiqi Fan, Fajie Yuan, Huaizong Shen

Nature Methods, 23(1), 88–95. DOI: [10.1038/s41592-025-02916-8](https://doi.org/10.1038/s41592-025-02916-8). Introduced Cryo-IEF, a self-supervised scientific vision foundation model that learns from noisy cryo-EM particle images to support automated protein and molecular-complex structure analysis.

Artificial intelligence foundation model automates cryo-EM structure determination 2026

Yang Yan, Huaizong Shen

Nature Methods Research Briefing, 23, 26–27. DOI: [10.1038/s41592-025-02917-7](https://doi.org/10.1038/s41592-025-02917-7).

CryoDECO: Deconstructing Extreme Compositional and Conformational Heterogeneity in Cryo-EM via Foundation Model Priors 2026

Yang Yan, Yanwanyu Xi, Shiqi Fan, Yifei Wang, Ziyun Tang, Fajie Yuan, Huaizong Shen

LangTaoSha Preprint Server. DOI: [10.65215/LTSpreprints.2025.12.30.000075](https://doi.org/10.65215/LTSpreprints.2025.12.30.000075). Prior-guided framework using foundation-model representations to analyze heterogeneous molecular mixtures and dynamic structural states.

Unsupervised cryo-EM images denoising and clustering based on deep convolutional autoencoder and k-means++ 2022

D. Zhang, Yang Yan, Y. Huang, B. Liu, Q. Zheng, J. Zhang, N. Xia

IEEE Transactions on Medical Imaging, 42(5), 1509–1521. Deep representation-learning framework for cryo-EM denoising and clustering.

Machine learning for structure determination in single-particle cryo-electron microscopy: A systematic review 2021

J. G. Wu, Yang Yan, D. X. Zhang, B. W. Liu, Q. B. Zheng, X. L. Xie, S. Q. Liu, S. X. Ge, et al.

IEEE Transactions on Neural Networks and Learning Systems, 33(2), 452–472. Review of ML methods for single-particle cryo-EM structure determination.

Selected earlier medical-imaging publications 2022–2023

S. Li, B. Liu, S. Li, X. Zhu, Yang Yan, D. Zhang; D. Zhang, B. Liu, Y. Huang, Yang Yan, et al.

Complex & Intelligent Systems, 2022; *Journal of Digital Imaging*, 2023. Deep-learning and ordinal-regression methods for medical image analysis.

Research Highlights

- **Large-scale scientific vision foundation model:** developed **Cryo-IEF**, the first foundation model for cryo-EM particle analysis, pretrained on **~65M noisy 2D molecular images**. The model learns transferable visual representations for classification, clustering, quality ranking, and 3D reconstruction, helping turn noisy microscope data into interpretable protein and molecular-complex structures with less manual expert tuning.
- **Automated scientific reconstruction system:** developed **CryoWizard**, an automated computational pipeline that integrates learned particle-quality priors into single-particle cryo-EM workflows, streamlining the path from raw microscope data to high-resolution 3D molecular structures.
- **Foundation-prior generative reconstruction:** developed **CryoDECO**, a prior-guided heterogeneous reconstruction framework that uses pretrained particle representations to improve robustness under compositional and conformational heterogeneity, where samples contain mixed molecular species and flexible structural motions.
- **Open-source scientific ML infrastructure:** created **cryodata**, a PyPI-installable data layer with **over 20.0k downloads** as of May 2026, converting raw cryo-EM software outputs into PyTorch-ready training and inference datasets with scalable preprocessing, LMDB storage, balanced sampling, and CryoSPARC/RELION interoperability.

Research Software and Open Source

2023–2026

Cryo-IEF: Scientific Vision Foundation Model for Cryo-EM Particle Analysis

<https://github.com/westlake-repl/Cryo-IEF>

Designed and released a self-supervised foundation-model ecosystem pretrained on ~65M cryo-EM particle images. Built PyTorch inference and downstream tooling for CryoRanker and CryoClustering, enabling learned representations to support structural classification, pose-aware clustering, particle-quality assessment, and automated reconstruction workflows.

2024–2026

CryoDECO: Foundation-Prior Heterogeneous Reconstruction

<https://github.com/yanyang1998/CryoDECO>

Developed a prior-guided reconstruction framework that uses Cryo-IEF representations to reduce dependence on random ab initio initialization and disentangle compositional classification from 3D reconstruction. The method targets compositional and conformational heterogeneity, where samples contain mixed molecular species and flexible structural motions that conventional pipelines require extensive expert tuning to analyze.

2024–2025

CryoWizard: Automated Scientific Reconstruction Pipeline

<https://github.com/SMART-StructBio-AI/CryoWizard>

Built and extended an end-to-end computational pipeline integrating CryoRanker with CryoSPARC, streamlining the path from raw microscope data to high-resolution 3D molecular structures with minimal manual intervention through command-line, web, and Chrome-extension interfaces.

2023–2026

cryodata: Open-Source Data Infrastructure for Scientific ML

<https://github.com/SMART-StructBio-AI/cryoief-data>

Created the reusable data layer used by Cryo-IEF, CryoDECO, and CryoWizard to convert raw CryoSPARC particle outputs into PyTorch-ready training and inference datasets. The package supports MRC/MRCs preprocessing, LMDB-backed dataset creation, Fourier/Hartley feature generation, balanced sampling, PyTorch data loading, and CryoSPARC-to-RELION metadata conversion. Packaged and documented the toolkit for `pip install cryodata`, reaching **over 20.0k downloads** as of May 2026.

Technical Skills

Machine Learning

Self-supervised learning, representation learning, foundation models, computer vision, clustering, autoencoders, generative/reconstruction models, image denoising.

Scientific ML Systems

PyTorch training and inference, large-scale data preprocessing, multiprocessing data pipelines.

Structural Biology

Single-particle cryo-EM, cryo-ET, heterogeneous reconstruction, particle-quality assessment, CryoSPARC/RELION workflows, ChimeraX visualization.

Research Agenda

I aim to build AI systems that learn from large-scale scientific imaging data and reason over molecular structure, dynamics, and biological context. My near-term focus is representation-guided heterogeneous reconstruction for cryo-EM and cryo-ET, where foundation-model priors can help automate difficult expert-driven workflows. My long-term goal is AI-guided structural biology: models and agents that integrate visual evidence, molecular identity, conformational motion, and native cellular context to support discovery from complex biological mixtures.